OPERATING SYSTEM DESIGN

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PROCESS & THREAD MANAGMENT

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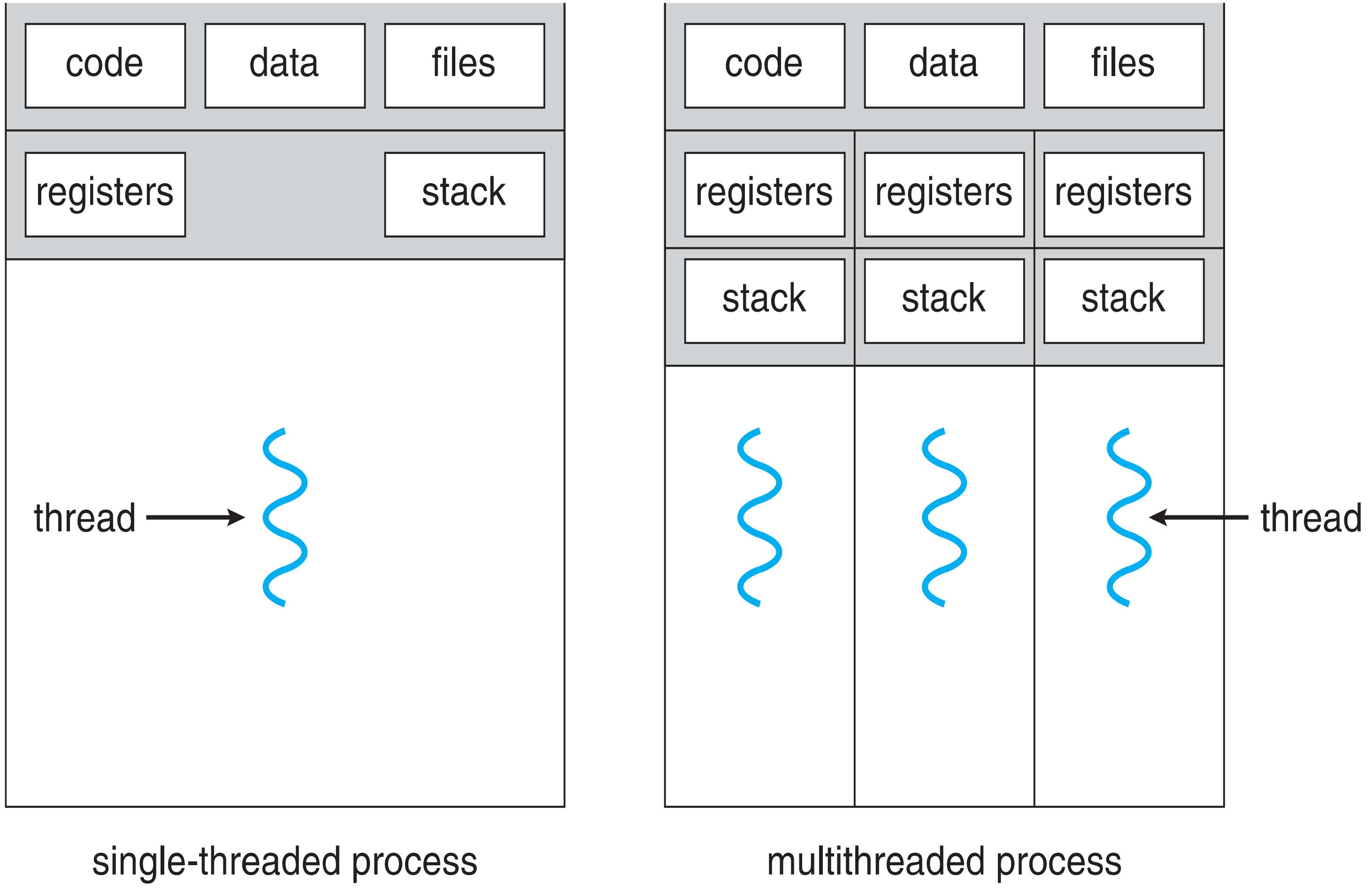
Multithreading Models

- Threads have different states (running, ready, Blocked, etc.)
- Save thread context, when not running
- Have private storage for local variables and execution stack
- Have shared access to the address space and resources (files etc.) of their process
  - when one thread alters (non-private) data, all other threads (of the process) can see this
  - threads communicate via shared variables
  - a file opened by one thread is available to others

Multithreading Models



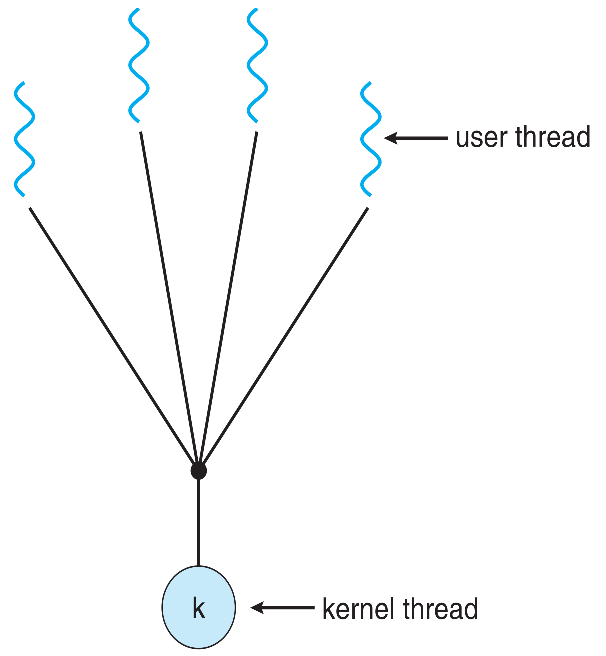
- **User threads** - management done by user-level threads library
- Three primary thread libraries:
  - POSIX **Pthreads**
  - Windows threads
  - Java threads
- **Kernel threads** - Supported by the Kernel
- Examples – virtually all general purpose operating systems, including:
  - Windows
  - Solaris
  - Linux
  - Tru64 UNIX
  - Mac OS X

Types

- Many-to-One
- One-to-One
- Many-to-Many

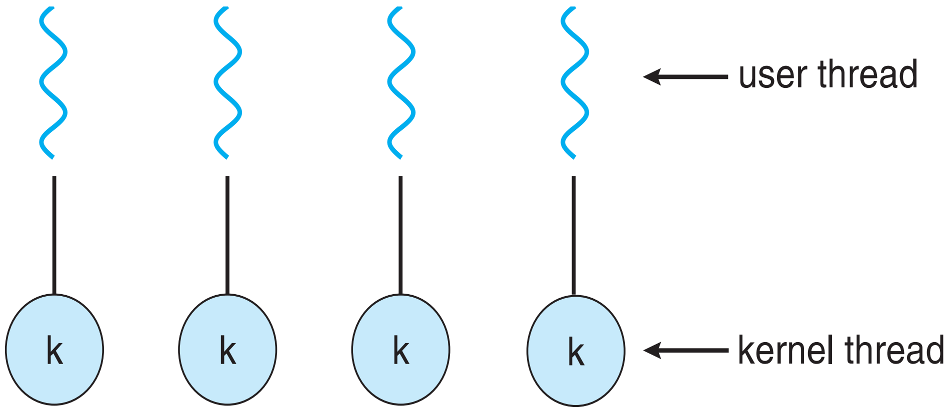
Many-to-One Model

- Many user-level threads mapped to single kernel thread
- One thread blocking causes all to block
- Multiple threads may not run in parallel on muticore system because only one may be in kernel at a time
- Few systems currently use this model
- Examples:
  - Solaris Green Threads
  - GNU Portable Threads



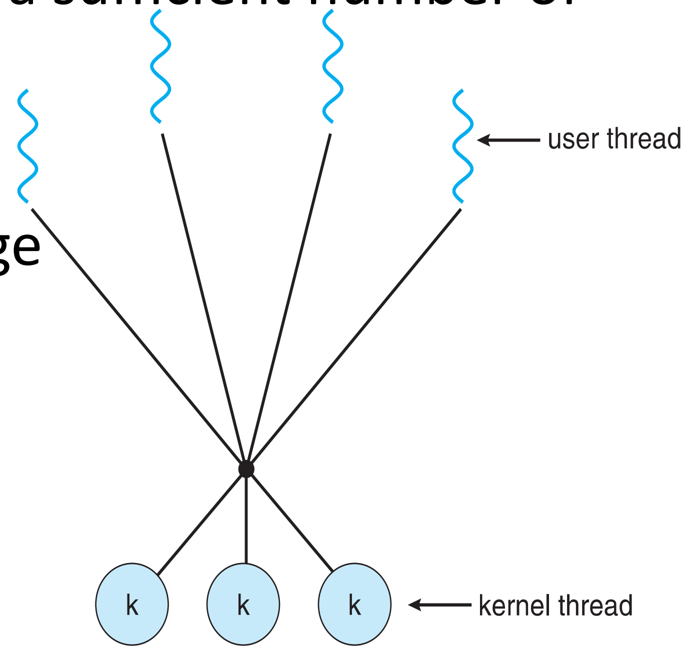
One-to-One Model

- Each user-level thread maps to kernel thread
- Creating a user-level thread creates a kernel thread
- More concurrency than many-to-one
- Number of threads per process sometimes restricted due to overhead
- Examples
  - Windows
  - Linux
  - Solaris 9 and later



Many-to-Many Model

- Allows many user level threads to be mapped to many kernel threads
- Allows the operating system to create a sufficient number of kernel threads
- Solaris prior to version 9
- Windows with the *ThreadFiber* package



THANK YOU

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